

Question Paper

Exam Date & Time: 01-Jan-2025 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH. (COMPUTER SCIENCE & ENGINEERING) EXAMINATIONS - DECEMBER 2024 / JANUARY 2025
SUBJECT: CSE 3123/CSE_3123 - OPERATING SYSTEMS

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Explain the six major categories of system calls in an operating system. (3)
- 1B) Relate how the degree of multi-programming is affected by long-term and medium-term scheduler. (3)
- 1C) Utilize the following page reference string: (4)
1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6.

Identify the number of page faults would occur for the following replacement algorithms, assuming four frames.

- _ FIFO replacement
_ Optimal replacement

- 2A) Consider the following system. (4)

Process	Arrival Time	Burst Time (ms)	Priority
P1	0	7	5
P2	2	5	7
P3	5	4	2
P4	3	6	2

The above system schedules processes using priority preemption with aging. Aging is a concept where priority (integer value) of a process is reduced by 1 after a certain time waiting in the ready queue. In the above system, the lower integer value represents higher priority. The aging time for the system is 3 ms.

Make use of Gantt chart,

- i) to schedule the processes as per the above-mentioned requirements.
ii) to calculate turnaround time and waiting time for each process.

- 2B) Consider the following snapshot of a system. (3)

Process	Allocation					Max					Available				
	A	B	C	D	E	A	B	C	D	E	A	B	C	D	E
P0	1	0	2	1	1	1	1	2	1	3	0	0	x	1	1
P1	2	0	1	1	0	2	2	2	1	0					
P2	1	1	0	1	0	2	1	3	1	0					
P3	1	1	1	1	0	1	1	2	2	1					

Utilize Banker's algorithm to find: (i) the content of the matrix Need. (ii) the smallest value of x for which the system in a safe state.

- 2C) Develop a process P_i that can implement mutual exclusion with test_and_set() instruction. Identify (3)

how the developed process supports mutual exclusion.

- 3A) i) Describe how swap space affects the performance of the operating system and where they reside. Illustrate it with a UNIX system example. (5)
ii) Assume that a disk drive has cylinders numbered 0 to 4999. Cylinder 143 is currently serving by drive, and the cylinder 125 was a previous request. The 86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130 are pending requests. What will be the time taken to service each request using SSTF algorithm.
(3+2 = 5 marks)
- 3B) i) What are the different Accessing Methods of a File? (5)
ii) Explain what is symbolic link and list at least two of its drawbacks.
(3+2 = 5 marks)
- 4A) Assuming a page size of 2K and a 32-bit machine, how many bits are used for the page number, and how many are used for the offset within the page? Give three main reasons why page sizes are usually powers of two. (4)
- 4B) How should the OS proceed if the processor wants to execute page 7, assuming that pages 1 to 5 of the active process are in main memory and pages 6 to 10 of the same process are stored in secondary memory? Explain with the block diagram. (4)
- 4C) Differentiate between logical address space and physical address space with an example. (2)
- 5A) A key design decision in OS memory management is the choice between paging and segmentation. Compare and contrast these two approaches to memory management, making sure to identify the strengths and weaknesses of each. (4)
- 5B) Explain locality model used to prevent thrashing. (3)
- 5C) Discuss the strengths and weaknesses of implementing an access matrix using access lists that are associated with objects. (3)

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